Self-Adaptive Improved Fungal Growth Optimization: A Hybrid Metaheuristic Framework for High-Quality, Robust and Scalable Clustering

Kalyan Kumar Das[1,2*], Souvik Sarkar Jr.[1], Anik Majumdar[1], Anirban Mukhopadhyay[2]

[1*]Department of Computer Science & Engineering, Techno India University, Kolkata, 700091, West Bengal, India.
[2]Department of Computer Science & Engineering, University of Kalyani, Nadia, West Bengal, India.

*Corresponding author(s). E-mail(s): kalyan.biltu@gmail.com;
Contributing authors: souviksarkarjr@gmail.com;
anikmajumdar61@gmail.com; anirban@klyuniv.ac.in;

Abstract

This study presents a Self-Adaptive Improved Fungal Growth Optimization (SAIFGO) technique, an improved version of the nature-inspired Fungal Growth Optimization (FGO) algorithm. The proposed method copies fungal behaviors such as hyphal growth, chemotropism, branching and spore germination. SAIFGO incorporates chaotic initialization, Levy flight-based mutation, opposition-based learning and an adaptive exploration–exploitation mechanism. A composite multi-objective fitness function is implemented using multiple cluster validity indices to extract improved clustering performance. The proposed algorithm is also hybridized with traditional clustering methods like K-Means and Fuzzy C-Means (FCM) for enhancement of centroid optimization. In addition, SAIFGO is evaluated against several metaheuristic algorithms using five clustering validity indices and multiple statistical analyses. SAIFGO outperforms or is competitive with all of the algorithms chosen for the experimental study. This implies that SAIFGO is a high-performing optimizer and a powerful alternative technique for clustering related complex optimization problems.

Keywords: K-Means, FCM, Fungal Growth Optimization (FGO), Self-Adaptive Improved Fungal Growth Optimization (SAIFGO)

1 Introduction

Clustering [1] is a fundamental unsupervised learning method for identifying inherent structures present within unlabeled data. It is a vital component in various applications, including pattern recognition [2], image segmentation [3] and data analysis where it simplifies complex data to facilitate efficient decision-making. However, the traditional clustering methods [4] including K-Means and Fuzzy C-Means (FCM), possess certain disadvantages. They are extremely sensitive to the initialization of cluster centroids, which often leads to premature convergence to local minima. In addition, the assumption of spherical cluster structures is considered one of the major drawbacks of these classical methods. The requirement of a fixed number of clusters limits their applicability. The pre-defined number of clusters might result in an incorrect representation of the data.

In order to solve these issues, researchers have developed new hybrid methods by integrating the use of meta-heuristic [5] optimization methods and conventional clustering techniques. Amongst all such hybrid methods, Fungal Growth Optimization (FGO) [6] technique exhibits promising results due to the inherent adaptive search nature of the algorithm. Nevertheless, there are various disadvantages with the traditional FGO method.

To overcome these shortcomings, the present research study aims at introducing a new form of FGO method called Self-Adaptive Improved Fungal Growth Optimization (SAIFGO).

The main contributions of this research can be summarized as follows:

- An improved balance between exploration and exploitation phases.
- The integration of chaotic initialization, Lévy flight-based mutation and opposition-based learning to enhance global search capability and convergence stability.
- The development of a composite multi-objective fitness function that combines multiple cluster validity indices for robust and balanced clustering.
- A novel hybrid clustering framework that integrates SAIFGO with K-Means and FCM for effective centroid refinement.
- Extensive experimental and statistical validation on benchmark datasets demonstrating the superiority of the proposed method.

The remaining part of this article is organized as follows. Section 2 discusses related work. Section 3 presents the proposed algorithm in detail. Experimental results on multiple datasets are presented in Section 4 along with statistical analysis. In Section 5 real-life applications of the proposed SAIFGO algorithm are shown. Finally, Section 6 concludes the findings.

2 Related Work

Cluster analysis has been extensively used for different applications in machine learning [7] and data mining [8]. Among all clustering techniques, K-Means [9] and Fuzzy C-Means [10] clustering are the most popular and computationally efficient techniques.

These conventional methods exhibit notable shortcomings. For instance, the initial values of parameters and local optimum convergence are major limitations of the mentioned clustering techniques.

Different metaheuristic techniques have been integrated with these traditional clustering techniques to overcome the limitations. The metaheuristic techniques selected for this study include : Genetic Algorithm Optimization (GA) [11], Particle Swarm Optimization (PSO) [12], Gorilla Troop Optimization (GTO) [13], Billiards Optimization (BO) [14], Improved Social Group Optimization (ISGO) [15], Energy Optimization (EO) [16], Hippo Optimization (HO) [17], Meta-OOD [18], Sequential Pattern Mining Optimization (SPM) [19] and General Trend (GT) [20]. The existing metaheuristic-based hybrid clustering algorithms are facing major challenges like low convergence rate, high computational complexity and exploration-exploitation balance despite of the improvements.

A relatively new algorithm, Fungal Growth Optimization (FGO) has shown promising results. Fungal Growth Optimization is a nature-inspired metaheuristic algorithm that emulates fundamental fungal growth mechanisms including hyphal-growth, chemotropism, branching, and spore germination. Hyphal-growth replicated hyphal extension to precisely explore the search space. Chemotropism helps to reach and exploit nutrient rich regions inside the search space, promoting performance against stagnation into local optima and slow convergence speed. Branching behavior replicates the way how new hyphal branches form next to an existing hypha explore regions for searching of more nutrient. Branching behavior promotes the exploratory operator. Germination represents how existing hyphae explore new environment to reach safer and nutrient rich areas. Due to its adaptive exploration capability and progressive refinement of candidate solutions, Fungal Growth Optimization (FGO) has demonstrated improved reliability compared to many conventional metaheuristic approaches.

To further enhance its performance, several variants of FGO have been introduced in recent studies. For instance, Bioluminescent Fungi Optimization (BFO) [21] incorporates a light-based attraction mechanism that enhances population diversity and improves the exploration capability of the algorithm. Similarly, Fungi Kingdom Expansion (FKE) [22] employs deterministic expansion strategies, combining immobile expansion for local search and mobile expansion for global search, thereby achieving a more balanced exploration–exploitation trade-off.

FGO still suffers from limitations such as high computational complexity, sensitivity to parameter settings and slow convergence in high-dimensional problems, which restrict its effectiveness for large-scale real life applications. Furthermore, adaptive parameter tuning [23], opposition-based learning [24], Lévy flight strategies [25] and hybridization [26] with traditional algorithms have been explored to improve the overall performance of FGO. These enhancements are introduced in SAIFGO for strengthening the balance between exploration and exploitation, accelerating convergence speed and reducing computational overhead.

3 Proposed Algorithm

3.1 Algorithm

Algorithm 1 Self-Adaptive Improved FGO Based Clustering (SAIFGO)

Require: Dataset $X = x_i{}_{i=1}^{N}$, number of clusters K, population size P, maximum iterations T
Ensure: Optimal centroids Z^*
Initialize bounds lb, ub from dataset
Generate chaotic sequence using logistic map: $r_{t+1} = 4r_t(1 - r_t)$
for $i = 1$ to P **do**
    Initialize fungi position $Z_{i,j} = lb + r_t(ub - lb)$
end for
Evaluate fitness $F(Z_i)$ using proxy objective
for $t = 1$ to T **do**
    Assign data points to nearest centroids $\ell_i = \arg\min_j \|x_i - c_j\|_2$
    Compute composite fitness $F = S + \frac{1}{1+DB} + \frac{CH}{1+CH} + D + \frac{1}{1+XB}$
    Update best solution Z_{best}
    Compute adaptive coefficient $r(t) = r_{\max} - \left(\frac{t}{T}\right)^{\beta} (r_{\max} - r_{\min})$
    for each fungi Z_i **do**
        Update position $Z_i^{t+1} = Z_i^t + r(t)(Z_{\text{best}}^t - Z_i^t)$
        Apply Lévy perturbation $Z_i^{t+1} = Z_i^{t+1} + \alpha \cdot Levy(\lambda)$
        Apply Opposition-Based Learning $Z_i^{new} = lb + ub - Z_i$
        Apply boundary control
    end for
    Local refinement
    Update centroids using K-Means $c_j = \frac{1}{|C_j|} \sum_{x_i \in C_j} x_i$
    Update centroids using FCM $c_j = \frac{\sum_{i=1}^{N} u_{ij}^m x_i}{\sum_{i=1}^{N} u_{ij}^m}$
end for
return Z_{best}

Algorithm 1 illustrates the workflow of the proposed SAIFGO algorithm.

3.2 Mathematical Formulation

3.2.1 Problem Definition

Let the dataset be represented as:

$$X = \{x_1, x_2, \ldots, x_N\}, \quad x_i \in \mathbb{R}^d \tag{1}$$

where N denotes the total number of data samples and d represents the feature dimensionality.

The clustering objective is to partition X into K disjoint clusters by determining optimal centroids:

$$C = \{c_1, c_2, \ldots, c_K\}, \quad c_j \in \mathbb{R}^d \tag{2}$$

In the proposed Improved-FGO framework, each candidate solution encodes all cluster centroids as:

$$Z_i = [c_1, c_2, \ldots, c_K] \in \mathbb{R}^{K \times d} \tag{3}$$

The clustering task is therefore formulated as a constrained optimization problem:

$$\max_{Z} \; F(Z) \tag{4}$$

subject to:

$$c_j \in [lb, ub]^d \tag{5}$$

where lb and ub denote lower and upper bounds derived from the dataset feature space.

3.2.2 Cluster Assignment Mechanism

For a given solution Z, each data sample is assigned to its nearest centroid using Euclidean distance formula:

$$\ell_i = \arg \min_{j \in \{1,\ldots,K\}} \|x_i - c_j\|_2 \tag{6}$$

This assignment produces K clusters $C_1, C_2, \ldots, C_K$.

3.2.3 Composite Fitness Function

To enhance clustering robustness and structural validation, a multi-objective fitness function is constructed by integrating five validity indices:

$$F = S + \frac{1}{1+DB} + \frac{CH}{1+CH} + D + \frac{1}{1+XB} \tag{7}$$

Here, DB and XB are transformed into maximization form.

For intermediate iterations, a computationally efficient proxy objective is employed:

$$F = -\sum_{i=1}^{N} \min_{j} \|x_i - c_j\| \tag{8}$$

To ensure compatibility, minimization-based indices (DB and XB) are transformed into maximization form using inverse normalization. Equal weighting is adopted to maintain balance among different criteria.

3.2.4 Silhouette Index

$$S = \frac{1}{N} \sum_{i=1}^{N} \frac{b(i) - a(i)}{\max(a(i),\, b(i))} \tag{9}$$

where $a(i)$ denotes the intra-cluster distance and $b(i)$ represents the nearest-cluster distance.

3.2.5 Davies–Bouldin Index

$$DB = \frac{1}{K} \sum_{i=1}^{K} \max_{j \neq i} \left(\frac{\sigma_i + \sigma_j}{\|c_i - c_j\|} \right) \tag{10}$$

where σ_i is the average intra-cluster dispersion of cluster i.

3.2.6 Calinski–Harabasz Index

$$CH = \frac{\mathrm{Tr}(B_K)/(K-1)}{\mathrm{Tr}(W_K)/(N-K)} \tag{11}$$

where B_K and W_K represent the between-cluster and within-cluster scatter matrices, respectively.

3.2.7 Dunn Index

$$D = \frac{\min_{i \neq j} \delta(C_i, C_j)}{\max_k \Delta(C_k)} \tag{12}$$

where $\delta(C_i, C_j)$ denotes the inter-cluster distance and $\Delta(C_k)$ represents the cluster diameter.

3.2.8 Xie–Beni Index

$$XB = \frac{\sum_{j=1}^{K} \sum_{x_i \in C_j} \|x_i - c_j\|^2}{N \cdot \min_{i \neq j} \|c_i - c_j\|^2} \tag{13}$$

3.2.9 Chaotic Initialization

Population diversity is enhanced using logistic chaotic mapping:

$$r_{t+1} = 4 r_t (1 - r_t) \tag{14}$$

Initial positions are generated as:

$$Z_{i,j} = lb + r_t (ub - lb) \tag{15}$$

3.2.10 Adaptive Position Update Strategy

To balance exploration and exploitation, an adaptive coefficient is defined as:

$$r(t) = r_{\max} - \left(\frac{t}{T}\right)^{\beta} (r_{\max} - r_{\min}) \tag{16}$$

where T denotes the maximum number of iterations.

The fungi position update rule is expressed as:

$$Z_i^{(t+1)} = Z_i^t + r(t) \left(Z_{\text{best}}^t - Z_i^t\right) \tag{17}$$

3.2.11 Lévy Flight Exploration

To prevent premature convergence, Lévy perturbation is introduced:

$$Z_i^{(t+1)} = Z_i^{(t+1)} + \alpha \cdot \mathrm{Levy}(\lambda) \tag{18}$$

3.2.12 Opposition-Based Learning

For poorly performing solutions, opposition learning is applied:

$$Z_i^{\text{new}} = lb + ub - Z_i \tag{19}$$

3.2.13 K-Means Update

$$c_j = \frac{1}{|C_j|} \sum_{x_i \in C_j} x_i \tag{20}$$

3.2.14 Fuzzy C-Means Update

Membership:

$$u_{ij} = \frac{\left(\frac{1}{\|x_i - c_j\|}\right)^{\frac{2}{m-1}}}{\sum_{k=1}^{K} \left(\frac{1}{\|x_i - c_k\|}\right)^{\frac{2}{m-1}}} \tag{21}$$

Centroid update:

$$c_j = \frac{\sum_{i=1}^{N} u_{ij}^m x_i}{\sum_{i=1}^{N} u_{ij}^m} \tag{22}$$

3.2.15 Final Optimization Model

The overall SAIFGO based clustering model is defined as:

$$\max_{Z \in [lb,\, ub]^{K \times d}} F(Z) \tag{23}$$

3.3 Flow Chart

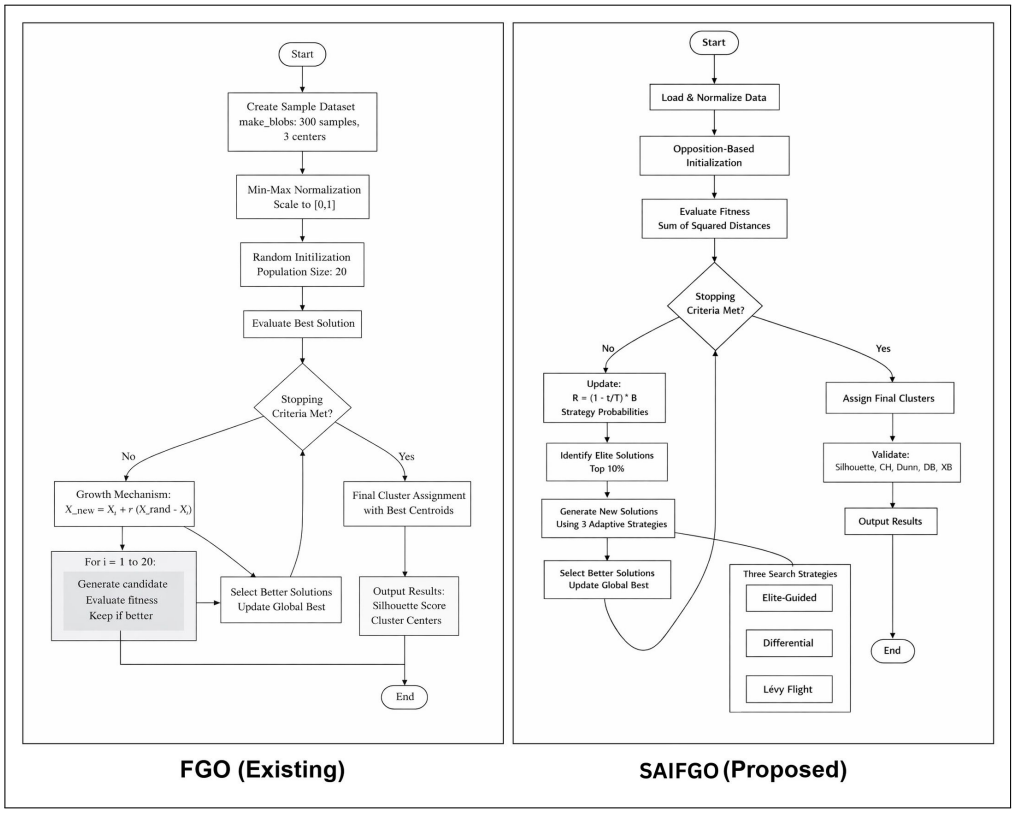


Fig. 1 Comparative Flowchart of FGO and SAIFGO Algorithms

Figure 1 displays a comparative flowchart of the traditional FGO and the proposed SAIFGO algorithm. It highlights their initialization, iterative improvement process, clustering steps and final result generation. SAIFGO contains structural enhancements and better solution-updating mechanisms. SAIFGO introduces several key enhancements: it utilizes opposition-based initialization rather than random selection, incorporates a triple-strategy update mechanism, and identifies the top elite solutions to improve convergence. Furthermore, SAIFGO adds a multi-metric validation phase to ensure the robustness of the final clusters before outputting results.

Table 1 Comparison of SAIFGO with other state-of-the-art Metaheuristic Algorithms

Aspects	GA	PSO	FGO	SAIFGO (Proposed)
Mutation Strategy	Random mutation	No mutation	No mutation	Lévy flight mutation
Population Initialization	Random	Random position and velocity	Random	Opposition-based learning
Initialization Formula	Random initialization	$X_i \sim U(lb, ub)$, $V_i \sim U(-v_{max}, v_{max})$	$X_i \sim U(lb, ub)$	$X_{\text{opposite}} = lb + ub - X$
Exploration Mechanism	Crossover	Velocity update ($pbest$, $gbest$)	Random interaction	$(X_j - X_k)$ search
Strategy Adaptation	Fitness-based selection	Inertia weight adjustment	Fixed strategy	Adaptive strategy

Table 1 showcases the difference between the FGO and the proposed SAIFGO algorithm. It highlights the key modifications introduced in the proposed SAIFGO.

3.4 Novel Contributions of the Proposed SAIFGO Framework

The proposed Self-Adaptive Improved Fungal Growth Optimization (SAIFGO) algorithm introduces a structurally enhanced and self-adaptive optimization framework for clustering. Unlike conventional metaheuristic modifications that rely on isolated improvements, SAIFGO integrates multiple mechanisms within a unified adaptive search paradigm. The key novel contributions are summarized as follows:

1. **Self-Adaptive Multi-Strategy Optimization Mechanism:**
   A dynamic strategy selection framework is introduced in which multiple search operators—elite-guided exploitation, differential exploration, and Lévy-based mutation—compete during the optimization process. Instead of using fixed probabilities, SAIFGO employs a success-based adaptation mechanism that continuously updates the selection probability of each strategy according to its historical performance. This enables automatic prioritization of effective search behaviors, suppression of underperforming operators, and data-driven adaptation of the search process, transforming the optimizer into a self-learning system.
2. **Elite-Mean Guided Search for Stable Convergence:**
   Unlike other traditional metaheuristics that are greatly affected by the best global solution, SAIFGO uses the elite mean guidance strategy to determine its search direction based on the mean of the best-performing solutions. This minimizes the effects of noise-affected solutions and maximizes the diversity of solutions.
3. **Strategy-Level Adaptive Exploration–Exploitation Control:**
   SAIFGO strikes a balance between exploration and exploitation through the adaptive switching of search behaviors. Elite-guided updates facilitate exploitation, differential interactions facilitate exploration, and Lévy flights facilitate jumps. The switching of search behaviors enables the adaptation of the optimization process based on the state of the search.
4. **Closed-Loop Feedback-Driven Optimization:**

A feedback loop between the quality of solutions and the behavior of the search process is established, where improvements in fitness directly affect the selection of future strategies. This creates a form of reinforcement learning, which continually enhances the adaptability of the search process in complex optimization landscapes.

5. **Diversity-Preserving Initialization with Opposition-Based Sampling:**
   SAIFGO also includes a better initialization approach that involves a combination of random population generation with opposition-based solutions. This approach for sampling the population space is likely to improve the chances of exploring better areas in the space during the initial iterations.
6. **Composite Multi-Objective Fitness Formulation for Clustering:**
   A single fitness function is constructed using a combination of different cluster validity measures, i.e., Silhouette, Davies-Bouldin, Calinski-Harabasz, Dunn, and Xie-Beni. This helps in optimizing all aspects of cluster compactness, separation, and integrity.
7. **Hybrid Optimization–Refinement Framework:**
   The proposed SAIFGO is combined with K-Means and Fuzzy C-Means (FCM) algorithms to create a new hybrid model. The global optimization for the position of centroids is performed by the proposed SAIFGO algorithm and local search is performed by using the K-Means and Fuzzy C-Means algorithms.

Overall, the proposed SAIFGO algorithm framework introduces a self-adaptive, feedback-driven, multi-strategy optimization paradigm that significantly enhances clustering performance, convergence stability and robustness across diverse datasets.

Table 2 Comparison of Optimization Features Across Metaheuristic Algorithms

Algorithm	Lévy	OBL	Multi-objective	Hybrid	Self-adaptive	Adaptive
GA	×	×	×	×	×	×
PSO	×	×	×	✓	×	Partial
GTO	×	×	×	✓	×	✓
BO	×	×	×	×	×	✓
ISGO	×	✓	×	✓	✓	✓
EO	×	×	×	×	×	✓
HO	×	×	×	✓	✓	✓
FGO	×	×	×	×	×	×
SPM	×	×	✓	✓	×	✓
Meta-OOD	×	×	✓	✓	✓	✓
General Trend	✓	✓	✓	✓	✓	✓
SAIFGO (Proposed)	✓	✓	✓	✓	✓	✓

3.5 Computational Complexity Analysis

The computational complexity of the proposed Self-Adaptive Improved Fungal Growth Optimization (SAIFGO) algorithm depends on the population size (P), number of iterations (T), dataset size (N), number of clusters (K) and feature dimension (d).

In each iteration the dominant computational cost arises from the cluster assignment process where each data point is assigned to its nearest centroid. This requires computing distances between N data points and K centroids resulting in a complexity of $\mathcal{O}(NK)$ per candidate solution. Considering P candidate solutions over T iterations the overall time complexity of the algorithm can be expressed as:

$$\mathcal{O}(PTNK) \tag{24}$$

Additional operations such as Lévy flight-based mutation and opposition-based learning introduce an extra computational cost of $\mathcal{O}(P)$ per iteration. However, this cost is relatively negligible compared to the dominant clustering computation.

In comparison, traditional clustering algorithms such as K-Means and Fuzzy C-Means (FCM) exhibit a time complexity of $\mathcal{O}(NKT)$. The higher computational cost of SAIFGO is attributed to its population-based optimization framework. SAIFGO provides improved global search capability, better avoidance of local optima and enhanced clustering performance.

The space complexity of the proposed algorithm is determined by the storage of candidate solutions and is given by :

$$\mathcal{O}(PKd) \tag{25}$$

4 Results and Discussion

The results are presented here. The following cluster validity indices and statistical analyses are used to compare the quality of all formed clusters.

Silhouette Score [27] measures the degree to which each data point fits within its assigned cluster. High value represents better separation and cohesion between the formed clusters.

Davies-Bouldin Index [28] evaluates the mean similarity between clusters. Less value represents clustering with more distinct groups.

Dunn Index [29] evaluates cluster separation and compactness. High value indicates well-separated clusters.

Xie-Beni Index [30] assesses compactness and separation. Low value represents well-defined clusters.

Caliński-Harabasz Index [31] is used to determine the optimal number of clusters. Higher value shows better results.

Table 3 Description of the datasets used in the experimental study

Dataset	Domain	Samples	Features	Clusters(optimal)	Availability
Iris	Botany	150	6	3	UCI_ML Repo
Drug	Healthcare	1,000	9	3	UCI_ML Repo
Bank	Finance	2,512	16	3	UCI_ML Repo
Shopping	E-Commerce	3,900	18	4	UCI_ML Repo
Wave	Signal Processing	12,762	3	3	UCI_ML Repo
Placement	Education	10,000	10	2	Kaggle
Delivery	Supply Chain	25,000	15	5	Kaggle

Table 3 summarizes the domain, number of samples, feature counts, optimal number of clusters, and availability for each dataset used in the study.

4.1 Comparison of Clustering Validity Indices

Table 4 Comparison of clustering validity indices on the Bank Dataset

Algorithm	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
SAIFGO+K-Means	0.280317242	0.318054925	919.9743706	0.022517772	1.428217924
SAIFGO+FCM	0.276692507	0.535316698	911.9646896	0.022517772	1.456834273
FGO+K-Means	0.250692	0.784791	723.976616	0.01901	1.323301
FGO+FCM	0.200912	3.822037	488.605095	0.009445	1.900755
K-Means	0.250600	0.784498	723.959000	0.01900	1.323900
FCM	0.200800	3.908100	488.709500	0.00940	1.900743
GA+K-Means	0.250731	0.785700	723.974949	0.01901	1.323019
GA+FCM	0.200803	3.869657	488.509370	0.009445	1.899800
PSO+K-Means	0.250692	0.784791	723.976616	0.01901	1.323301
PSO+FCM	0.200923	3.820879	488.669571	0.009445	1.900030
GTO+K-Means	0.250692	0.784791	723.976616	0.01901	1.323301
GTO+FCM	0.200864	3.881321	488.437263	0.009445	1.901460
BO+K-Means	0.250692	0.784791	723.976616	0.01901	1.323301
BO+FCM	0.200803	3.850107	488.509370	0.009445	1.900743
ISGO+K-Means	0.250692	0.784791	723.976616	0.01901	1.323301
ISGO+FCM	0.200864	3.880952	488.437263	0.009445	1.901460
EO+K-Means	0.250692	0.784791	723.976616	0.01901	1.323301
EO+FCM	0.200864	3.880117	488.437263	0.009445	1.901460
HO+K-Means	0.250692	0.784791	723.976616	0.01901	1.323301
HO+FCM	0.200991	3.821126	488.649606	0.009445	1.900537
MetaOOD+K-Means	0.237700	1.898700	87.365900	0.44510	2.366400
MetaOOD+FCM	0.008736	112.3308	0.336310	0.33631	1.320500
SPM+K-Means	0.068500	5.946900	163.623400	0.33290	4.425900
SPM+FCM	0.068300	14.61490	162.701700	0.33290	6.494400
GT+K-Means	0.083100	3.367100	195.803900	0.40650	2.932700
GT+FCM	0.058300	60.30980	150.601300	0.33620	11.98870

Table 4 showcases the comparison of clustering validity indices on the Bank Dataset. The SAIFGO integrated with K-Means outperforms other K-Means based hybrid algorithms, achieving the highest Silhouette Score, CH-Score and Dunn-Index, as well as the lowest XB-Index. These results suggest that the SAIFGO+K-Means approach provides superior cluster separation and compactness for this specific

dataset. Similarly, the SAIFGO+FCM configuration demonstrates superior performance among the fuzzy-logic-based approaches.

Table 5 Comparison of clustering validity indices on the Delivery Dataset

Algorithm	**Silhouette**	**XB-Index**	**CH-Score**	**Dunn-Index**	**DB-Index**
SAIFGO+K-Means	0.216181980	0.489172921	7841.937318	0.007251419	1.652433130
SAIFGO+FCM	0.139232180	0.941437248	6782.355194	0.004275072	2.247269050
FGO+K-Means	0.221523	0.860723	8132.152518	0.009185	1.627833
FGO+FCM	0.141532	1.433497	6931.204013	0.006222	2.215477
K-Means	0.221500	0.860600	8132.150400	0.009200	1.628000
FCM	0.141600	1.440400	6931.580100	0.006200	2.215500
GA+K-Means	0.221523	0.860723	8132.152518	0.009185	0.009185
GA+FCM	0.141540	1.433522	6931.208537	0.006222	1.627833
PSO+K-Means	0.221523	0.860723	8132.152518	0.009185	1.627833
PSO+FCM	0.141548	1.433481	6931.114820	0.006222	2.215711
GTO+K-Means	0.221523	0.860723	8132.152518	0.009185	1.627833
GTO+FCM	0.141531	1.433343	6931.141342	0.006222	2.215685
BO+K-Means	0.221523	0.860723	8132.152518	0.009185	1.627833
BO+FCM	0.141532	1.433507	6931.204013	0.006222	2.215477
ISGO+K-Means	0.221523	0.860723	8132.152518	0.009185	1.627833
ISGO+FCM	0.141531	1.433552	6931.141342	0.006222	2.215685
EO+K-Means	0.221523	0.860723	8132.152518	0.009185	1.627833
EO+FCM	0.141547	1.433487	6931.080421	0.006222	2.215774
HO+K-Means	0.221523	0.860723	8132.152518	0.009185	1.627833
HO+FCM	0.141525	1.433530	6931.147928	0.006222	2.215446
MetaOOD+K-Means	0.115200	1.653900	3392.630600	0.148300	2.416200
MetaOOD+FCM	0.009639	1824.6965	0.121288	0.121288	1.244700
SPM+K-Means	0.079600	3.872600	2344.094100	0.104900	3.725200
SPM+FCM	0.072600	6.729600	2319.622200	0.065100	4.773600
GT+K-Means	0.095800	3.358600	2964.650700	3.364200	3.364200
GT+FCM	0.046800	12.97650	2216.141900	0.102100	6.558400

Table 5 signifies that for the Delivery Dataset, K-Means based hybrid algorithms generally demonstrate superior performance compared to FCM based approaches across most validity indices. While standard K-Means and traditional metaheuristic hybrids provide highly competitive Silhouette and CH-Scores, the SAIFGO+K-Means algorithm achieves the most optimal XB-Index. This indicates that the proposed SAIFGO optimization effectively enhances cluster compactness and separation beyond the capabilities of standard initialization, even in cases where the dataset strongly favors a K-Means centroid structure.

Table 6 Comparison of clustering validity indices on the Drug Dataset

Algorithm	S-Score	XB-Index	CH-Index	Dunn-Index	DB-Index
SAIFGO+K-Means	0.248218602	0.360244726	391.0712996	0.027000671	1.394291987
SAIFGO+FCM	0.250264287	0.493596964	388.7150453	0.032327321	1.389608321
FGO+K-Means	0.212309	0.780998	255.813186	0.040165	1.600545
FGO+FCM	0.203297	0.963209	239.140871	0.026304	1.578206
K-Means	0.212300	0.781000	255.813200	0.040200	1.600500
FCM	0.202500	0.967000	238.309000	0.026300	1.577400
GA+K-Means	0.212309	0.780998	255.813186	0.040165	1.600545
GA+FCM	0.203297	0.963212	239.140871	0.026304	1.578206
PSO+K-Means	0.212309	0.780998	255.813186	0.040165	1.600545
PSO+FCM	0.203297	0.963209	239.140871	0.026304	1.578206
GTO+K-Means	0.212309	0.780998	255.813186	0.040165	1.600545
GTO+FCM	0.203297	0.963209	239.140871	0.026304	1.578206
BO+K-Means	0.212309	0.780998	255.813186	0.040165	1.600545
BO+FCM	0.203297	0.962145	239.140871	0.026304	1.578206
ISGO+K-Means	0.212309	0.780998	255.813186	0.040165	1.600545
ISGO+FCM	0.203297	0.963212	239.140871	0.026304	1.578206
EO+K-Means	0.212309	0.780998	255.813186	0.040165	1.600545
EO+FCM	0.203297	0.963209	239.140871	0.026304	1.578206
HO+K-Means	0.212309	0.780998	255.813186	0.040165	1.600545
HO+FCM	0.203297	0.963209	239.140871	0.026304	1.578206
MetaOOD+K-Means	0.084300	1.797500	93.970300	0.216200	2.658000
MetaOOD+FCM	0.073950	388.4845	0.216341	0.216341	3.899200
SPM+K-Means	0.089200	3.105900	99.434300	0.219300	3.215300
SPM+FCM	0.087200	8.016300	98.540400	0.216500	4.857200
GT+K-Means	0.090900	2.170200	99.928700	0.226500	2.815400
GT+FCM	0.085800	6.615200	94.558200	0.208100	5.084000

Table 6 demonstrates that hybrid algorithms integrated with the SAIFGO optimization significantly outperform all other evaluated models on the Drug Dataset. Notably, the SAIFGO+FCM configuration achieves the highest overall Silhouette Score and the most optimal DB-Index, indicating exceptionally well-defined cluster boundaries. Simultaneously, SAIFGO+K-Means delivers superior results in cluster compactness and variance-based structure, producing the highest CH-Index and the lowest XB-Index. These results confirm that the SAIFGO algorithm substantially enhances clustering quality, providing a more robust separation between groups.

Table 7 Comparison of clustering validity indices on the Iris Dataset

Algorithm	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
SAIFGO+K-Means	0.520362149	0.110525328	365.4529675	0.133525211	0.765732140
SAIFGO+FCM	0.520362149	0.114563191	365.4529675	0.133525211	0.765732140
FGO+K-Means	0.452950	0.304155	237.842018	0.082819	0.896584
FGO+FCM	0.452319	0.230330	237.673998	0.082819	0.895600
K-Means	0.452900	0.304200	237.842000	0.082800	0.896600
FCM	0.452300	0.230600	237.674000	0.082800	0.895600
GA+K-Means	0.452950	0.304155	237.842018	0.082819	0.896584
GA+FCM	0.452319	0.230600	237.673998	0.082819	0.896584
PSO+K-Means	0.452950	0.304155	237.842018	0.082819	0.896584
PSO+FCM	0.452319	0.230308	237.673998	0.082819	0.895600
GTO+K-Means	0.452950	0.304155	237.842018	0.082819	0.896584
GTO+FCM	0.452319	0.230335	237.673998	0.082819	0.895600
BO+K-Means	0.452950	0.304155	237.842018	0.082819	0.896584
BO+FCM	0.452319	0.230343	237.673998	0.082819	0.895600
ISGO+K-Means	0.452950	0.304155	237.842018	0.082819	0.896584
ISGO+FCM	0.452319	0.230408	237.673998	0.082819	0.895600
EO+K-Means	0.452950	0.304155	237.842018	0.082819	0.896584
EO+FCM	0.452319	0.230421	237.673998	0.082819	0.895600
HO+K-Means	0.452950	0.304155	237.842018	0.082819	0.896584
HO+FCM	0.452319	0.230419	237.673998	0.082819	0.895600
MetaOOD+K-Means	0.459600	0.733400	136.361300	0.079100	0.732700
MetaOOD+FCM	0.522900	0.234100	267.507200	0.202000	0.800000
SPM+K-Means	0.519200	0.678300	265.724100	0.111800	1.453300
SPM+FCM	0.519200	0.739300	265.724100	0.111800	1.504800
GT+K-Means	0.500200	0.462700	255.415400	0.082800	1.166000
GT+FCM	0.500200	0.499600	255.415400	0.082800	1.204600

Table 7 displays that for the Iris Dataset, the hybrid algorithms based on the SAIFGO provide the overall highest Silhouette Score, CH-Score and Dunn-Index while achieving the lowest XB-Index. Furthermore it is observed that other traditional metaheuristic-based hybrid algorithms yield nearly identical clustering validity indices. This suggests that while standard metaheuristics converge to similar local optima, the SAIFGO successfully explores a more optimal solution space resulting in superior cluster separation and structural compactness.

Table 8 Comparison of clustering validity indices on the Placement Dataset

Algorithm	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
SAIFGO+K-Means	0.139579935	0.754380292	1726.659193	0.027293046	2.072542825
SAIFGO+FCM	0.172074591	13879.10083	2241.978226	0.029666586	2.065217362
FGO+K-Means	0.143576	1.551743	1873.041327	0.016483	2.160249
FGO+FCM	0.060793	262.153398	1465.828933	0.029239	3.430847
K-Means	0.143500	1.551900	1873.035100	0.016500	2.160249
FCM	0.061000	264.199400	1465.967300	0.029200	3.431200
GA+K-Means	0.143576	1.551743	1873.041327	0.016483	2.160249
GA+FCM	0.060958	264.187475	1465.967348	0.029239	3.431195
PSO+K-Means	0.143576	1.551743	1873.041327	0.016483	2.160249
PSO+FCM	0.060793	262.091415	1465.828933	0.029239	3.430847
GTO+K-Means	0.143576	1.551743	1873.041327	0.016483	2.160249
GTO+FCM	0.060958	264.232995	1465.967348	0.029239	3.431195
BO+K-Means	0.143576	1.551743	1873.041327	0.016483	2.160249
BO+FCM	0.060958	264.491498	1465.967348	0.029239	3.431195
ISGO+K-Means	0.143576	1.551743	1873.041327	0.016483	2.160249
ISGO+FCM	0.060958	264.359163	1465.967348	0.029239	3.431195
EO+K-Means	0.143576	1.551743	1873.041327	0.016483	2.160249
EO+FCM	0.060958	264.232699	1465.967348	0.029239	3.431195
HO+K-Means	0.143576	1.551743	1873.041327	0.016483	2.160249
HO+FCM	0.060957	264.823884	1465.804091	0.029239	3.431841
MetaOOD+K-Means	0.123100	1.764900	1444.395800	0.064700	2.415400
MetaOOD+FCM	0.044168	773.4710	1041.063200	0.035815	7.574300
SPM+K-Means	0.115100	2.419200	1436.636400	0.057600	2.777500
SPM+FCM	0.055500	37.12180	1212.148400	0.057600	9.866300
GT+K-Means	0.114300	2.338400	1425.839100	0.057600	2.708200
GT+FCM	0.043900	368.6133	1136.767400	0.035800	27.88240

Table 8 illustrates a relatively low Silhouette Score across all evaluated hybrid algorithms, indicating that the clustering structure within the Placement Dataset is inherently weak or highly overlapped. Despite these structural challenges the SAIFGO provides a moderate enhancement to both K-Means and FCM performance, specifically achieving the highest Silhouette Score and CH-Score with the FCM variant. The exceptionally high XB-Index values observed across several models (including the fuzzy-based configurations), suggest a degree of instability or high sensitivity to centroid placement. These findings underscore the relative robustness of the SAIFGO optimization in extracting meaningful patterns where standard approaches struggle.

Table 9 Comparison of clustering validity indices on the Shopping Dataset

Algorithm	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
SAIFGO+K-Means	0.139579935	0.754380292	1726.659193	0.027293046	2.072542825
SAIFGO+FCM	0.082093056	10876.78463	440.7948008	0.021867151	2.179250546
FGO+K-Means	0.138171	1.033940	635.142676	0.038299	1.971172
FGO+FCM	0.130630	283058.9835	596.841064	0.032192	2.031137
K-Means	0.138300	1.042200	634.735800	0.022800	1.976400
FCM	0.126600	619807.3775	582.165300	0.026200	2.051000
GA+K-Means	0.138171	1.033940	635.142676	0.038299	1.971172
GA+FCM	0.130828	315311.0058	594.080995	0.038622	2.044415
PSO+K-Means	0.138171	1.033940	635.142670	0.038299	1.971172
PSO+FCM	0.130404	263630.8936	594.344436	0.027982	2.029268
GTO+K-Means	0.138171	1.033940	635.142676	0.038299	1.971172
GTO+FCM	0.130790	261871.5754	591.964607	0.035306	2.040989
BO+K-Means	0.138171	1.033940	635.142676	0.038299	1.971172
BO+FCM	0.130809	272393.7638	594.784106	0.025731	2.040332
ISGO+K-Means	0.138171	1.033940	635.142676	0.038299	1.971172
ISGO+FCM	0.131314	284890.5852	594.239679	0.021474	2.033080
EO+K-Means	0.138171	1.033940	635.142676	0.038299	1.971172
EO+FCM	0.131560	276193.9413	598.118716	0.037344	2.035323
HO+K-Means	0.138171	1.033940	635.142676	0.038299	1.971172
HO+FCM	0.131437	260650.5586	596.383926	0.028656	2.038899
MetaOOD+K-Means	0.090700	3.894600	390.946600	0.352500	3.331000
MetaOOD+FCM	0.136720	2199426.08	565.382000	0.419841	8.774600
SPM+K-Means	0.067300	11.67670	378.963400	0.324100	6.610700
SPM+FCM	0.064800	30.61360	374.134900	0.327500	9.762500
GT+K-Means	0.076800	4.286500	372.201400	0.273900	4.058100
GT+FCM	0.054300	16318668.6	292.103700	0.259158	4.466900

Table 9 denotes that for the Shopping Dataset, FCM-based hybrid algorithms demonstrate significant instability, as evidenced by extreme XB-index values and lower CH-scores, which indicate poor cluster compactness and substantial overlap. Among all evaluated approaches, SAIFGO+K-Means provides the most balanced performance by successfully optimizing the cluster structure. It achieves the highest CH-Score and the most optimal XB-Index effectively mitigating the structural weaknesses inherent in the dataset. These results suggest that the SAIFGO optimization offers a more stable and reliable configuration for centroid-based clustering.

Table 10 Comparison of clustering validity indices on the Wave Dataset

Algorithm	Silhouette	XB-Index	CH-Score	Dunn-Index	DB-Index
SAIFGO+K-Means	0.531432048	0.114455905	17782.38721	0.005233699	0.735841602
SAIFGO+FCM	0.525469226	0.131566397	17585.58914	0.002502896	0.737368943
FGO+K-Means	0.463731	0.305391	11012.34305	0.002605	0.904901
FGO+FCM	0.462599	0.269499	11002.59207	0.001215	0.903548
K-Means	0.463700	0.305400	11012.34300	0.002600	0.904900
FCM	0.462600	0.269500	11002.59210	0.001200	0.903500
GA+K-Means	0.463731	0.305391	11012.34305	0.002605	0.904901
GA+FCM	0.462587	0.269504	11002.38502	0.001215	0.903535
PSO+K-Means	0.463731	0.305391	11012.34305	0.002605	0.904901
PSO+FCM	0.462587	0.269498	11002.38502	0.001215	0.903535
GTO+K-Means	0.463731	0.305391	11012.34305	0.002605	0.904901
GTO+FCM	0.462587	0.269503	11002.38502	0.001215	0.903535
BO+K-Means	0.463731	0.305391	11012.34305	0.002605	0.904901
BO+FCM	0.462587	0.269501	11002.38502	0.001215	0.903535
ISGO+K-Means	0.463731	0.305391	11012.34305	0.002605	0.904901
ISGO+FCM	0.462599	0.269499	11002.59207	0.001215	0.903548
EO+K-Means	0.463731	0.305391	11012.34305	0.002605	0.904901
EO+FCM	0.462599	0.269498	11002.59207	0.001215	0.903548
HO+K-Means	0.463731	0.305391	11012.34305	0.002605	0.904901
HO+FCM	0.462587	0.269499	11002.38502	0.001215	0.903535
MetaOOD+K-Means	0.381900	0.417300	9852.674000	0.002200	1.129100
MetaOOD+FCM	0.462600	0.425400	11002.59210	0.001200	1.014400
SPM+K-Means	0.467400	0.441800	10821.57880	0.264100	1.121600
SPM+FCM	0.466000	0.546100	10946.98100	0.002900	1.200800
GT+K-Means	0.463700	0.305400	11012.34300	0.002600	0.904900
GT+FCM	0.462600	0.425400	11002.59210	0.001200	1.014400

Table 10 reveals that the proposed SAIFGO significantly enhances clustering performance on the Wave Dataset, which possesses a strong intrinsic structure. A substantial improvement is observed across all clustering validity indices compared to conventional metaheuristic-based hybrid algorithms. Most notably, SAIFGO+K-Means nearly doubles the CH-Score of the standard K-Means approach, while simultaneously achieving the most optimal Silhouette Score and the lowest DB-Index. These results confirm that the SAIFGO optimization is exceptionally effective at capturing well-defined cluster boundaries and maintaining structural density in large-scale datasets with clear patterns.

4.2 Statistical Analysis

4.2.1 Boxplot

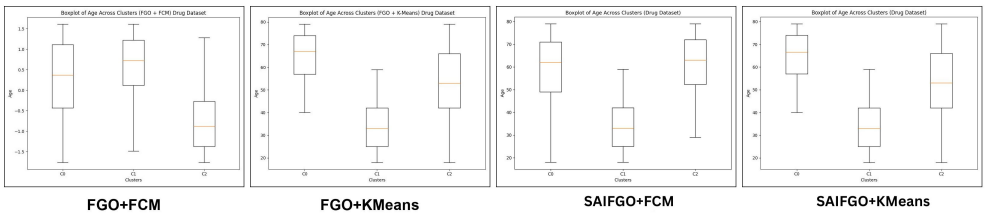


Fig. 2 Boxplot for the Drug Dataset

Figure 2 demonstrates that SAIFGO based variants achieve more distinct cluster separation compared to standard FGO variants. For the K-Means implementations, the SAIFGO clusters exhibit a notably reduced IQR. This indicates lower intra-cluster variability and higher consistency in age grouping. The improvement is most pronounced in the FCM variants. This results in minimal overlap between the distribution of age groups, suggesting that the SAIFGO optimization better captures the underlying data structure. The tighter whiskers and distinct median separation across both SAIFGO variants reinforce the statistical robustness and superior discriminative power of the proposed method on the Drug Dataset.

4.2.2 Coefficient of Variation

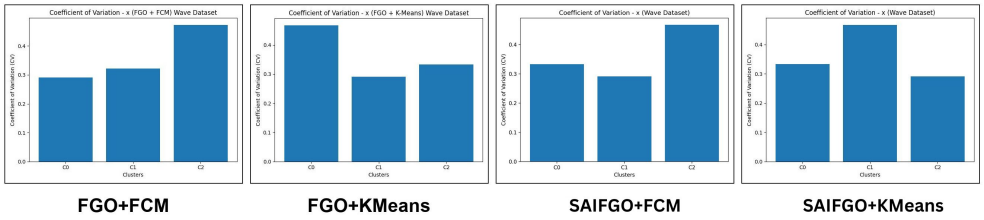


Fig. 3 Coefficient of Variation Analysis for the Wave Dataset

Figure 3 shows that SAIFGO based variants produce a more balanced and consistent clustering structure on the Wave Dataset compared to standard FGO. The Coefficient of Variation (CV) across clusters in SAIFGO variants shows a more uniform distribution of variability. For example, FGO+FCM exhibits a high CV in C2 (0.47) indicating a significant imbalance in cluster density compared to C0 (0.29). Similarly, FGO+KMeans shows noticeable variance with C0 reaching 0.46. In contrast, the

SAIFGO hybrids maintain a more controlled CV range. SAIFGO+FCM yields values of 0.33,0.29 and 0.46 ; while SAIFGO+KMeans achieves a similar balance with 0.33,0.46 and 0.29. This increased stability in centroid positioning and the avoidance of extreme CV fluctuations suggest that SAIFGO effectively optimizes the exploration–exploitation trade-off, allowing the algorithm to avoid local optima and achieve higher statistical consistency across all clusters.

4.2.3 Convergence Graph

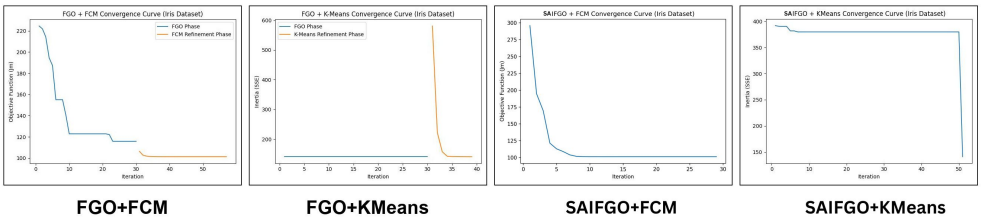


Fig. 4 Convergence Graph for the Iris Dataset

Figure 4 displays SAIFGO variants achieve superior optimization efficiency through faster convergence on the Iris Dataset. A clear distinction is visible in the FCM based hybrids. SAIFGO+FCM converges rapidly from an initial value of 295 to its minimum of approximately 100 within the first 10 iterations. In contrast, FGO+FCM exhibits a much slower, stepwise descent, requiring roughly 30 iterations to reach a similar objective value. This demonstrates SAIFGO's ability to locate high-quality search space regions more aggressively. Regarding the K-Means variants, SAIFGO+KMeans maintains a stable exploration phase before the refinement trigger, eventually achieving a competitive final inertia. The sharp vertical drops observed in both K-Means refinement phases signify the rapid local fine-tuning provided by the hybrid approach.

4.2.4 Friedman Test

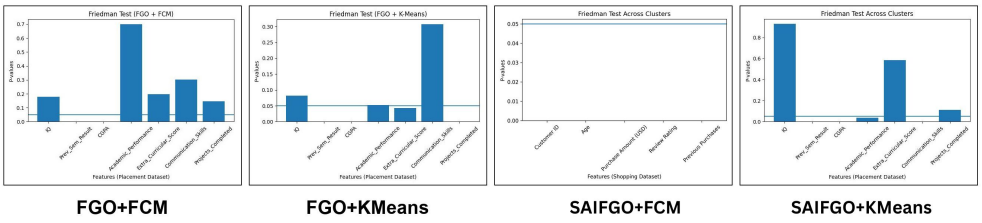


Fig. 5 Friedman Test for the Placement Dataset

Figure 5 reveals the statistical significance of features across the different clustering variants using the Friedman Test in the Placement Dataset. SAIFGO+FCM demonstrates exceptional statistical robustness, it achieves p-values of 0.00 across all features, including Age and Purchase Amount, indicating that the clusters are highly distinguishable. In contrast, the FGO variants show significant variability. In FGO+FCM, features like Academic Performance (p=0.70) and Communication Skills (p=0.30) fail to reach the 0.05 significance threshold, suggesting weaker discrimination. SAIFGO+KMeans shows improved significance for specific metrics like CGPA and Previous Semester Results compared to FGO+KMeans, which exhibits fluctuations in Communication Skills (p¿0.30). Overall, the SAIFGO based hybrids provide more consistent ranking and stronger statistical evidence for distinct cluster structures.

4.2.5 IQR Plot

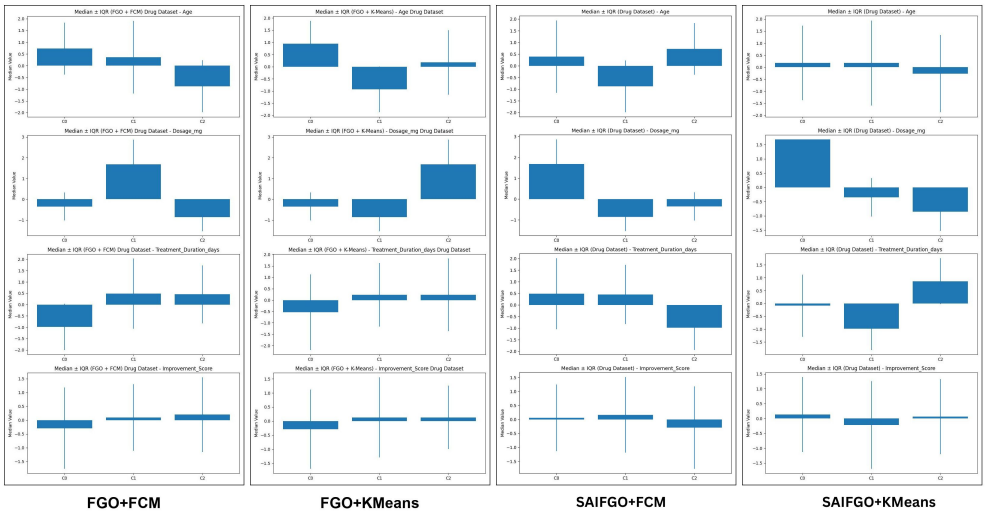


Fig. 6 IQR Plot for the Drug Dataset

Figure 6 indicates that SAIFGO based variants produce more distinctly spaced medians and a clearer cluster identity compared to FGO based algorithms. In the FGO variants several features particularly Dosage (mg) and Treatment Duration (days) show significant overlap in median ranges and wide IQRs which leads to ambiguity between C0,C1, and C2. The SAIFGO hybrids exhibit sharper separation. SAIFGO+KMeans, the median for Dosage (mg) at C0 reaches approximately 1.5, while C1 and C2 are clearly partitioned below zero. Furthermore, for Treatment Duration SAIFGO variants show significantly tighter IQRs, indicating lower intra-cluster variability. This distinct median separation and reduced overlap ensure better cluster inter-pretability reinforcing the statistical robustness and practical utility of the proposed optimization method.

4.2.6 Nemenyi Post-hoc Test

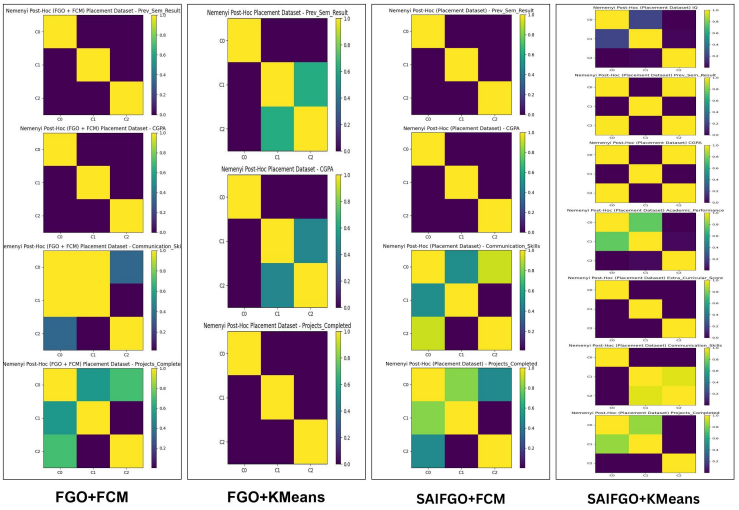


Fig. 7 Nemenyi Post-hoc Test for the Placement Dataset

Figure 7 exhibits that SAIFGO+FCM exhibits more distinct low p-value blocks (dark purple), indicating stronger cluster-to-cluster discrimination and fewer ambiguous comparisons than its FGO counterpart. The SAIFGO+KMeans variant demonstrates the most structured pairwise significance pattern, particularly for key features like Academic Performance and Projects Completed where the clear contrast between purple and yellow blocks confirms significant statistical differences between cluster pairs. FGO variants frequently display mid-range p-values (teal or, green zones between 0.3 and 0.7) suggesting a lack of clear separation in those specific features. The high concentration of extreme values with many comparisons approaching 0 and the diagonal consistently at 1 confirms that SAIFGO based hybrids provide more reliable and interpretable cluster pairs.

4.2.7 T-Test

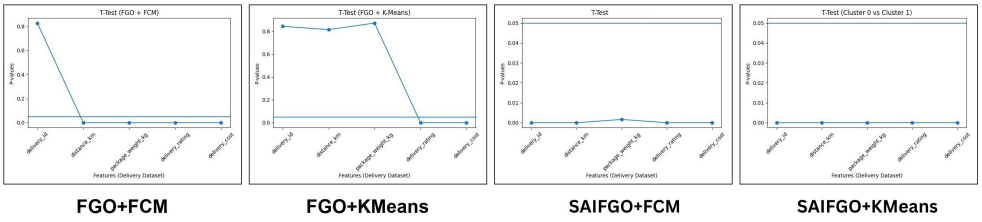


Fig. 8 T-Test for the Delivery Dataset

Figure 8 presents that SAIFGO based variants achieve significantly stronger mean differences between clusters compared to traditional FGO-based algorithms. In the FGO implementations, several dimensions exhibit a lack of statistical significance; specifically, FGO+FCM shows a high p-value for delivery_id (0.82), and FGO+KMeans demonstrates weak discrimination across delivery_id, distance_km (0.81), and package_weight_kg (0.87). These high p-values indicate that the clusters generated by FGO are not statistically distinguishable. Conversely, both SAIFGO+FCM and SAIFGO+KMeans produce p-values near zero across all evaluated features, staying well below 0.05 significance threshold. SAIFGO variants provide a more reliable and statistically robust segmentation for the Delivery Dataset.

4.2.8 Vargha-Delaney A12 Effect

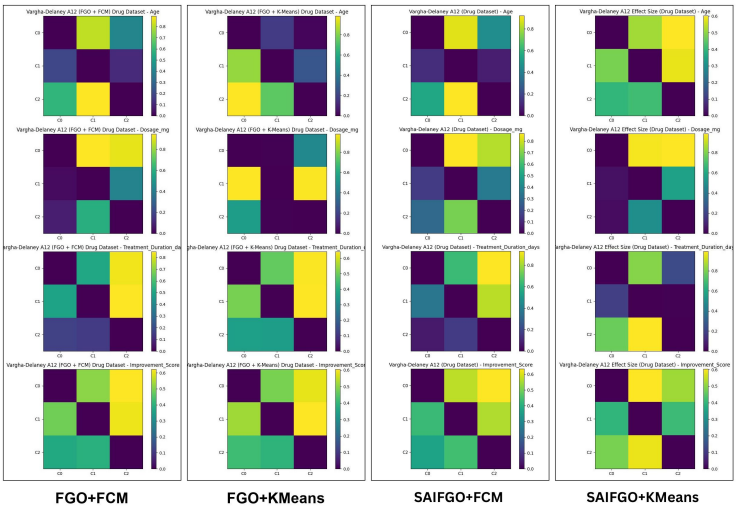


Fig. 9 Vargha–Delaney A12 Effect Size for the Delivery Dataset

Figure 9 indicates SAIFGO-based variants achieve a more pronounced distribution shift and higher practical significance compared to FGO-based algorithms on the Drug Dataset. heatmaps reveal that SAIFGO hybrids provide stronger cluster dominance, with values frequently reaching the 0.8–0.9 range (high dominance) or 0.1–0.2 range (low dominance). The FGO variants exhibit more moderate A12 values (0.3–0.7), particularly in features like Age and Improvement_Score, where many comparisons hover near the 0.5 mark. The presence of more extreme values and sharper color contrasts in the SAIFGO+FCM and SAIFGO+KMeans heatmaps confirms that these variants ensure more meaningful structural separation and practical robustness, resulting in clusters that are distributionally distinct across all evaluated features.

4.2.9 Wilcoxon Signed-Rank Test

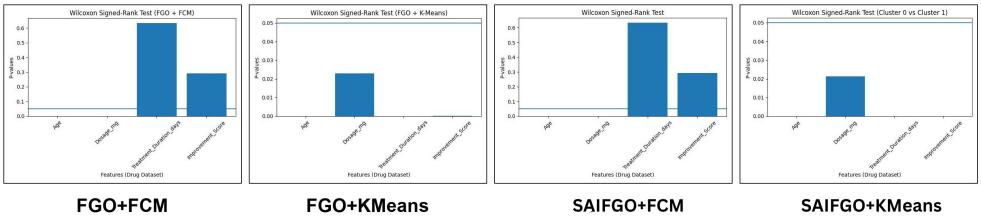


Fig. 10 Wilcoxon Signed-Rank Test for the Drug Dataset

Figure 10 suggests that in the FGO+FCM and SAIFGO+FCM implementations, certain variables like Treatment Duration (p=0.63) and Improvement Score (p=0.29) fail to cross the 0.05 significance threshold, indicating weaker sensitivity in those dimensions. However, SAIFGO+KMeans demonstrates significantly stronger statistical reliability, with p-values for all features falling below the significance line. Specifically, Dosage_mg is refined to p=0.02, while features like Age and Improvement_Score reach near 0.00; this improved sensitivity to variables within the Drug Dataset confirms that SAIFGO provides a higher confidence level for cluster differentiation.

5 Real-Life Application

5.1 Credit Card Anomaly Detection

5.1.1 Problem Statement

In modern time the usage of digital payments and online banking is increasing, along with it credit card fraud has become a major financial concern. These fraudulent transactions cause financial losses. Often the traditional rule-based algorithms are often unable to detect new sophisticated fraud patterns. There is a need for better algorithm that can automatically identify unusual and suspicious transactions.

5.1.2 Approach Used

The approach typically involves collecting transaction data and pre-processing it to remove noise and irrelevant information. SAIFGO+FCM provides more refined and adaptive cluster boundaries where, SAIFGO+KMeans provides better computational efficiency.

5.1.3 Data Source

Table 11 Description of the Datasets used in Credit Card Anomaly Detection

Dataset	Dimension	Description
Credit Card Dataset	$284,807 \times 31$	Provides transaction related information.

Table 11 summaries the datasets which are used to demonstrate the credit card anomaly detection real life application for the proposed algorithm. These datasets are available in the public domain, downloaded from the website Kaggle [32].

5.1.4 Application

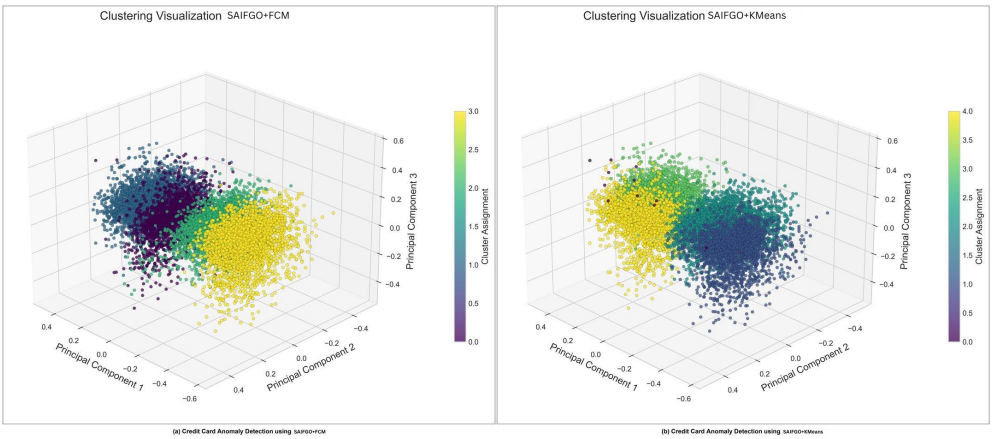


Fig. 11 Implementation for Detecting Unusual Activities in a Credit Card Dataset

Figure 11 shows the effectiveness of the SAIFGO-based hybrid approach to detect the abnormality of the credit card transactions. By leveraging fuzzy membership, SAIFGO+FCM algorithm forms more refined and adaptive cluster boundaries which enables better separation between the normal and the anomalous transactions comparatively. On the other hand SAIFGO+KMeans provides better speed of execution for large-scale credit card transaction analysis.

5.1.5 Analysis

For the Credit Card Dataset SAIFGO successfully optimizes cluster formation in both hybrid methods. FCM provides more flexible clustering, it is more useful for detecting uncertain patterns. Anomalous transactions typically appear near cluster boundaries, far from centroids or as sparsely distributed data points. K-Means based hybrid provides clearer cluster separation which is useful for identifying distinct transaction groups.

5.1.6 Insights

SAIFGO-based hybrid methods improve clustering performance by optimizing centroid initialization and by preventing algorithms from getting trapped in local minima. This leads to better cluster separation, more stable convergence and improved detection of anomalous transactions. As a result SAIFGO based hybrid algorithm provides more meaningful output.

5.2 Aviation Network

5.2.1 Problem Statement

India's domestic aviation network has a large number of airports which are interconnected through complex flight routes. Introducing a better clustering algorithm is a challenging task. Due to uneven geographical distribution along with varying passenger demand, regional connectivity gaps, and operational cost constraints one better algorithms is needed.

5.2.2 Approach Used

K-Means version of SAIFGO does hard clustering, each airport is assigned to one cluster. SAIFGO optimizes initial centroid selection to improve clustering stability and performance. SAIFGO integrated with FCM performs soft clustering. Airports can belong to multiple clusters with varying membership degrees. It enhances centroid positioning for better fuzzy partitioning.

5.2.3 Data Source

Table 12 Description of the Datasets used in Indian Aviation Network

Dataset	Dimension	Description
City Dataset	4231×4	Contains city-related information.
Flight Dataset	113339×11	Contains flight-related operational data.

Table 12 summaries the datasets which are used to demonstrate the aviation network real life application for the proposed algorithm. These datasets are available in the public domain, downloaded from the website Kaggle [32].

5.2.4 Application

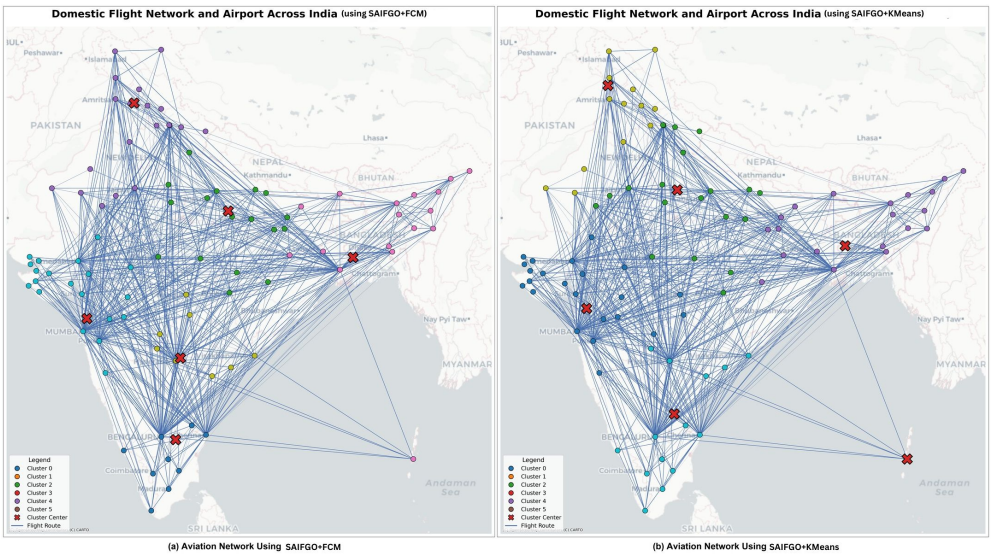


Fig. 12 Implementation on Indian Domestic Flight Dataset

Figure 12 displays that SAIFGO based hybrid algorithms produce clear cluster. Result shows natural regional divisions in domestic flight network along with strategic geographical positioning.

5.2.5 Analysis

West and South India show very dense route connections. It reflects high economic activity, greater airport concentration and strong domestic travel demand. The North-East airports with limited connectivity illustrate regional isolation due to geography. Interconnected routes demonstrates that Indian Domestic aviation is highly dense for which optimized hub and spoke management is needed. SAIFGO based algorithm effectively determine optimal hub location along with meaningful regional cluster which demonstrates practical applicability for the proposed algorithm.

5.2.6 Insights

SAIFGO introduces differential exploration and Lévy flight mutation which allows the algorithm to search many possible cluster center positions instead by avoiding the chances of getting stuck. In the aviation map, SAIFGO provides meaningful formed clusters by placing the centroids in the highly active areas. Traditional optimization algorithms can get stuck in local minima while doing clustering; on the other hand,

SAIFGO helps to escape poor solutions and refine cluster centers. The proposed algorithm shows enhanced exploration, mutation strategy and adaptive search resulting in more structured and meaningful airport network clusters.

6 Conclusion

The proposed Self-Adaptive Fungal Growth Optimization (SAIFGO) based hybrid clustering framework demonstrates consistency and robustness across diverse datasets. While achieving enhanced feature-level separation in high dimensional spaces, it maintains structural integrity (in low complexity scenarios) and variance stability. The approach effectively improves dispersion control, ensures reliable performance and reduces feature bias across different domains.

The outcome of the experiment proves the success of the proposed approach in providing high-quality clusters and high consistency. Incorporating the adaptive approach ensures that the problem of early convergence is avoided as the search space will be effectively explored.

It is evident that the presented approach can be scaled up and adapted for practical use. For increasing efficiency of the computational process while processing massive data sets, parallel algorithms may become the point of consideration in future research. Combining the presented technique with deep learning methods seems to be highly prospective.

In conclusion, the proposed SAIFGO-based clustering framework constitutes a reliable, dynamic and scalable solution to the contemporary problems of clustering; through improvement in search performance, convergence and quality of solutions, the method has greatly contributed towards the development of optimization-based unsupervised learning approaches.

Source Code and Data Availability

The SAIFGO algorithm implementation and associated datasets are publicly available to ensure transparency and reproducibility. All materials can be accessed through the following Google Drive link:

https://drive.google.com/drive/folders/1fEKb7RTR2LvLJiI5MT2vk8O9UPfBKL_P?usp=sharing

References

[1] Lin, L., Jia, T., Wang, Z., Zhao, J., Li, R.-H.: Psmc: Provable and scalable algorithms for motif conductance based graph clustering. In: Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, pp. 1793–1803 (2024)

[2] Djenouri, Y., Belhadi, A., Djenouri, D., Lin, J.C.-W.: Cluster-based information retrieval using pattern mining. Applied Intelligence **51**(4), 1888–1903 (2021)

[3] Naous, T., Sarkar, S., Abid, A., Zou, J.: Clustering plotted data by image segmentation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 21499–21504 (2022)

[4] Gionis, A., Mannila, H., Tsaparas, P.: Clustering aggregation. Acm transactions on knowledge discovery from data (tkdd) **1**(1), 4 (2007)

[5] Kurman, S., Kisan, S.: An in-depth and contrasting survey of meta-heuristic approaches with classical feature selection techniques specific to cervical cancer. Knowledge and Information Systems **65**(5), 1881–1934 (2023)

[6] Abdel-Basset, M., Mohamed, R., Abouhawwash, M.: Fungal growth optimizer: A novel nature-inspired metaheuristic algorithm for stochastic optimization. Computer Methods in Applied Mechanics and Engineering **437**, 117825 (2025)

[7] Jordan, M.I., Mitchell, T.M.: Machine learning: Trends, perspectives, and prospects. Science **349**(6245), 255–260 (2015)

[8] Wu, X., Kumar, V., Ross Quinlan, J., Ghosh, J., Yang, Q., Motoda, H., McLachlan, G.J., Ng, A., Liu, B., Yu, P.S., *et al.*: Top 10 algorithms in data mining. Knowledge and information systems **14**(1), 1–37 (2008)

[9] Ikotun, A.M., Ezugwu, A.E., Abualigah, L., Abuhaija, B., Heming, J.: K-means clustering algorithms: A comprehensive review, variants analysis, and advances in the era of big data. Information Sciences **622**, 178–210 (2023)

[10] Wu, C., Hou, J.: New semi-supervised fuzzy c-means clustering with asymmetric deviation constraints and fast algorithm. Expert Systems with Applications, 129648 (2025)

[11] Kouser, K., Priyam, A., Gupta, M., Kumar, S., Bhattacharjee, V.: Genetic algorithm-based optimization of clustering algorithms for the healthy aging dataset. Applied Sciences **14**(13), 5530 (2024)

[12] Abualigah, L.: Particle swarm optimization: advances, applications, and experimental insights. Computers, Materials, & Continua **82**(2), 1539 (2025)

[13] Abdollahzadeh, B., Gharehchopogh, F.S., Mirjalili, S.: Gorilla troops optimizer: A new nature-inspired metaheuristic algorithm for global optimization. Expert Systems with Applications **180**, 115016 (2021)

[14] Kaveh, A., Ghazaan, M.I.: Billiards optimization algorithm: A novel physics-based metaheuristic algorithm. Engineering with Computers (2023)

[15] Naik, A.: Enhanced social group optimization algorithm for solving optimization problems. Informatica **49**(1) (2025)

[16] Azizi, M., Aickelin, U., Khorshidi, H.A., Shishehgarkhaneh, M.B.: Energy-valley optimizer: A novel metaheuristic algorithm for global and engineering optimization. Scientific Reports **13**, 226 (2023)

[17] Amiri, M.H., Mehrabi Hashjin, N., Montazeri, M., Mirjalili, S., Khodadadi, N.: Hippopotamus optimization algorithm: A novel nature-inspired optimization algorithm. Scientific Reports **14**, 5032 (2024)

[18] Xu, G.: An adaptive parameter tuning of particle swarm optimization algorithm. Applied Mathematics and Computation **219**(9), 4560–4569 (2013)

[19] Dinh, T., Fournier-Viger, P., Hong, H.V.: Mining compact high utility sequential patterns. arXiv preprint arXiv:2302.11426 (2023)

[20] Decker, T., Koebler, A., Lebacher, M., Thon, I., Tresp, V., Buettner, F.: Explanatory model monitoring to understand the effects of feature shifts on performance. In: Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, pp. 550–561 (2024)

[21] Sabet-Rasekh, S., Beheshti, Z.: Bioluminescent fungi optimization algorithm and its applications to solve engineering optimization problems. Iran Journal of Computer Science **8**(4), 2307–2343 (2025)

[22] Alnahwi, F.M., Al-Yasir, Y.I., Sattar, D., Ali, R.S., See, C.H., Abd-Alhameed, R.A.: A new optimization algorithm based on the fungi kingdom expansion behavior for antenna applications. Electronics **10**(17), 2057 (2021)

[23] Wallingford, M., Li, H., Achille, A., Ravichandran, A., Fowlkes, C., Bhotika, R., Soatto, S.: Task adaptive parameter sharing for multi-task learning. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 7561–7570 (2022)

[24] Feurer, M., Klein, A., Eggensperger, K., Springenberg, J., Blum, M., Hutter, F.: Efficient and robust automated machine learning. ACM Transactions on Knowledge Discovery from Data (TKDD) **13**(2), 1–26 (2019)

[25] Alabed, T., Servi, S.: A lévy flight based chaotic black-winged kite algorithm for solving optimization problems. Scientific Reports **15**, 34608 (2025)

[26] Yan, M., Wang, Y., Pang, K., Xie, M., Li, J.: Efficient mixture of experts based on large language models for low-resource data preprocessing. In: Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, pp. 3690–3701 (2024)

[27] Rousseeuw, P.J.: Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. Journal of Computational and Applied Mathematics **20**, 53–65 (1987) https://doi.org/10.1016/0377-0427(87)90125-7

[28] Davies, D.L., Bouldin, D.W.: A cluster separation measure. IEEE Transactions on Pattern Analysis and Machine Intelligence **1**(2), 224–227 (1979) https://doi.org/10.1109/TPAMI.1979.4766909

[29] Dunn, J.C.: Well-separated clusters and optimal fuzzy partitions. Journal of Cybernetics **4**(1), 95–104 (1974) https://doi.org/10.1080/01969727408546059

[30] Xie, X.L., Beni, G.: A validity measure for fuzzy clustering. IEEE Transactions on Pattern Analysis and Machine Intelligence **13**(8), 841–847 (1991) https://doi.org/10.1109/34.85677

[31] Caliński, T., Harabasz, J.: A dendrite method for cluster analysis. Communications in Statistics - Theory and Methods **3**(1), 1–27 (1974) https://doi.org/10.1080/03610927408827101

[32] Kaggle: Kaggle: Your Machine Learning and Data Science Community. https://www.kaggle.com/. Accessed: 2026-04-01 (n.d.)